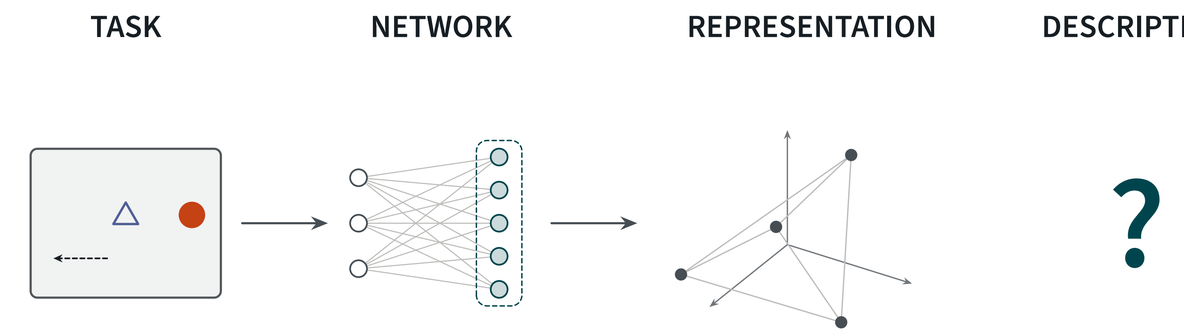
Decomposing representations through task symmetries

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SCAN FOR
supplement, proofs, references
esther-poniatowski.github.io/royaumont2026

1 PROBLEM
Understanding representations requires an interpretable description



How is the structure of a task reflected in the structure of representations?

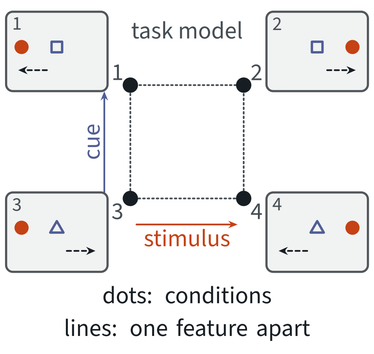
2 OBSERVATIONS
Common descriptors lack key properties

DESCRIPTOR & LIMITATION	PROPERTY NEEDED
Principal components	
⊗ meaning assigned afterwards	→ ✓ meaning fixed by the task
Trained decoders	
⊗ one variable at a time	→ ✓ whole representation
Decoding every split, or CCGP (Bernardi et al., 2020)	
⊗ many scores, or one summary	→ ✓ few components
ANOVA	
⊗ factorial designs only	→ ✓ various task designs

3 APPROACH
An answer combines distinct contributions from the researcher, mathematics and data

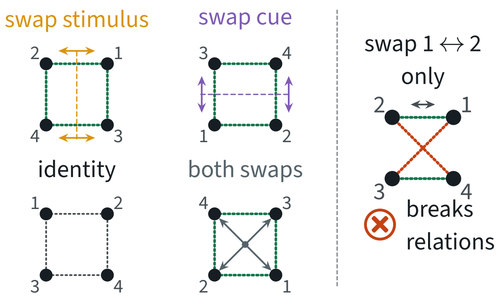
RESEARCHER *specifies* the task structure

1 Design the task



2 Specify the task symmetries

symmetry: manipulation of the conditions that preserves their relations

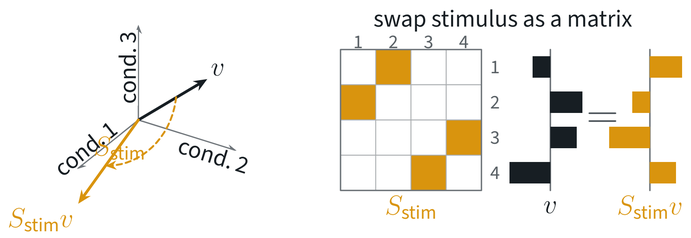


MATHEMATICS *determines* the task-based components

3 Encode symmetries in the condition space

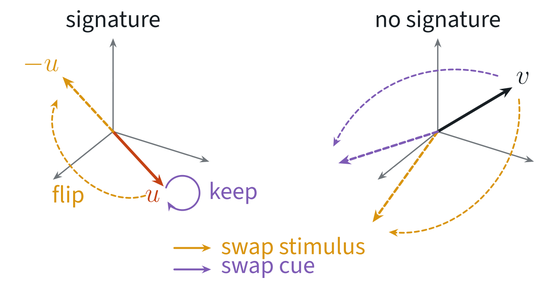
vector: one value per condition (e.g. tuning profile, similarity pattern, contrast)

symmetry: permutation of the condition values



4 Derive the isotypic components

isotypic component: patterns that respond to the task transformations according to a *signature* → reflects structure, not labels



RESEARCHER *reads* their meaning

DATA *estimate* the similarity matrix, the weights and the couplings

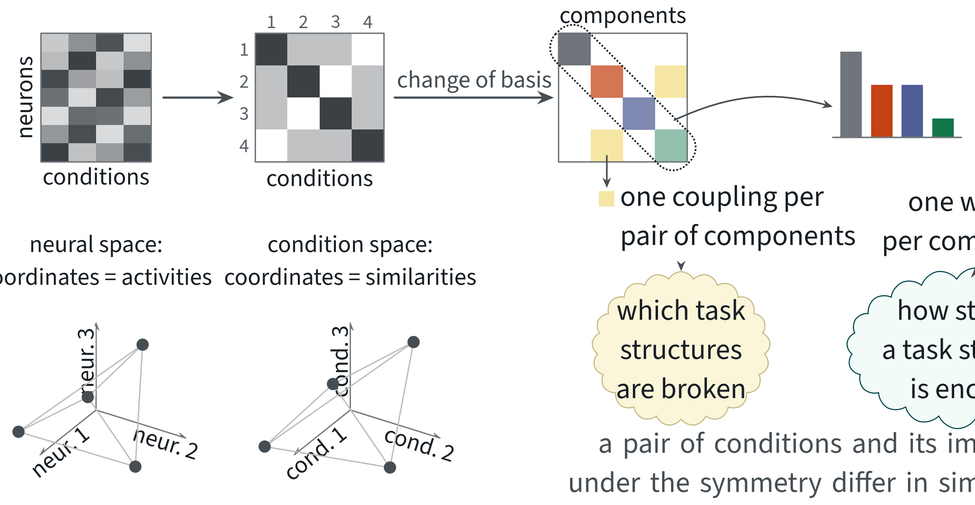
5 Interpret the components

meaning of a component: a task structure reflected by the symmetries that transform this component

symmetries (+: keep, −: flip)	mean	1	2	3
identity	+	+	+	+
swap stimulus	+	−	+	−
swap cue	+	+	−	−
both swaps	+	−	−	+
varies with	−	stimulus	cue	stimulus × cue
ANOVA correspondence	mean	main effect	main effect	interaction

components

6 Compute the similarity matrix



7 Extract the weights and couplings

one coupling per pair of components

one weight per component

which task structures are broken

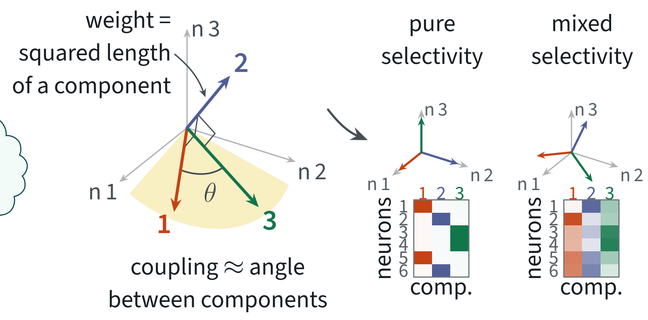
how strongly a task structure is encoded

a pair of conditions and its image under the symmetry differ in similarity

8 Map back in neural space

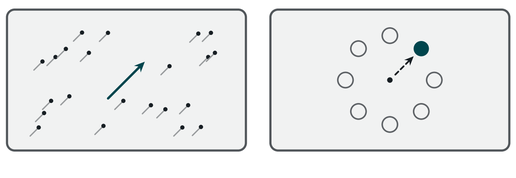
weights and couplings translate in geometry

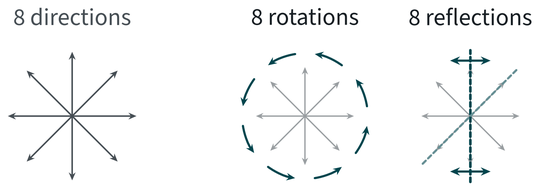
neural realizations share weights and couplings when they relate by rotation



4 BEYOND FACTORIAL DESIGNS
The same approach applies across task designs

EXPERIMENT motion discrimination — **TASK MODEL** — **SYMMETRIES**





STIMULUS: dots moving in one of 8 directions
RESPONSE: the target in that direction

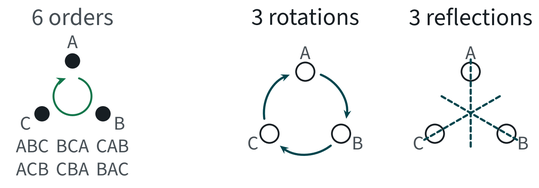
COMPONENTS Fourier modes

symmetries (+: keep, −: flip)	mean (dim 1)	360° (dim 2 · ♀ any phase)	180° (dim 2 · ♀ any phase)	120° (dim 2 · ♀ any phase)	90° (dim 1)
half-turn	+	−	+	−	+
quarter-turn	+	turns 90°	−	turns 270°	+
eighth-turn	+	turns 45°	turns 90°	turns 135°	−
reflection	+	mirror	mirror	mirror	±
varies with	−	direction	axis		cardinal or oblique

components

EXPERIMENT order memory — **TASK MODEL** — **SYMMETRIES**





STIMULUS: 3 locations flash in order (Xie et al., 2022)
RESPONSE: the same order, after a delay

COMPONENTS not directly inferred from any task variable

symmetries (+: keep, −: flip)	mean (dim 1)	chirality (dim 1)	item at rank (dim 4 = 2 × 2)
rotation	+	+	turns 120°
reflection	+	−	mirror
varies with	−	direction of the path	which location at which rank

components